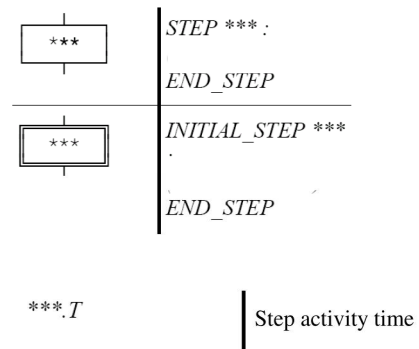
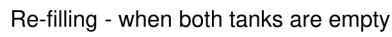
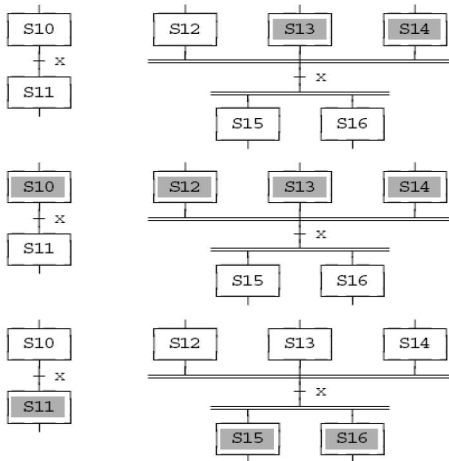




Principles of programming

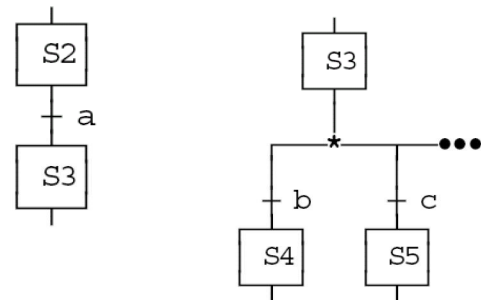
- The two steps cannot be directly connected - they must always be separated by a transition;
- Two transitions cannot be directly connected - they must always be separated by a step.



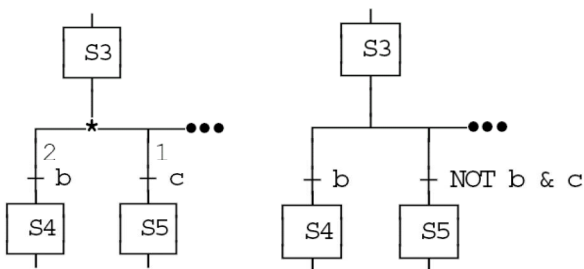
Rules for implementing the programme

- *Transit* is only *allowed* when all immediately preceding steps are *active*.
- If a transition is allowed and the associated transition condition is fulfilled at the same time, the *transition* is *cancelled* and all steps *immediately preceding* it are *deactivated* (or *reset*) and all steps *immediately following* the transition symbol are *activated*.
- This mode of operation, in which the attribute of an active state is passed to the next steps, resembles a *token passing* between successive stations in the network, and is sometimes also called a *token passing*.

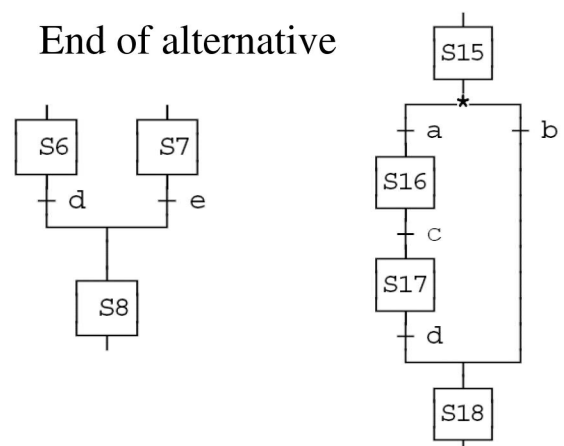
Structures: sequence and alternative



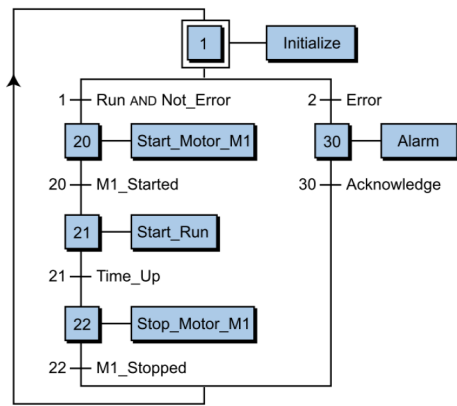
Alternative (options)



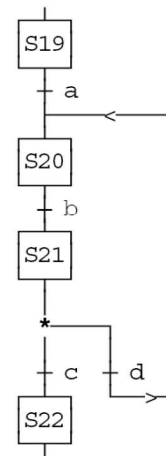
End of alternative



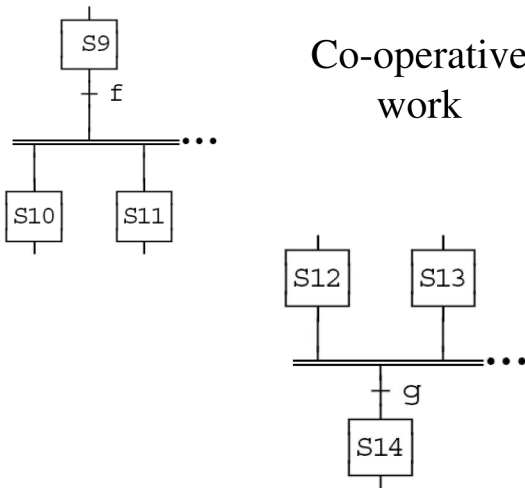
Fault handling



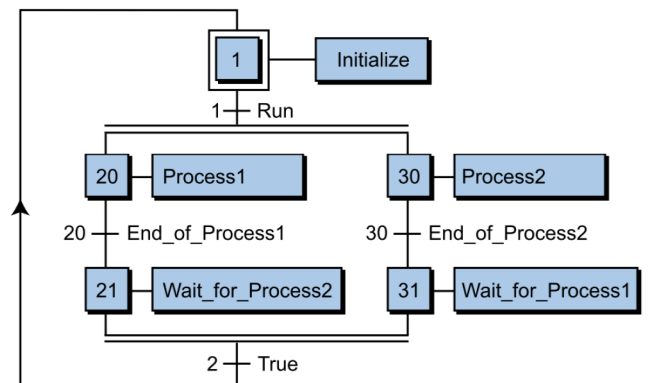
Loop



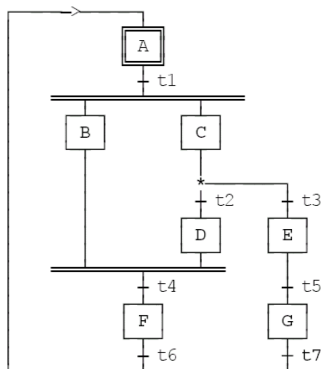
Co-operative work



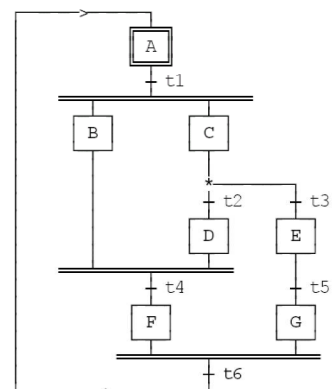
Synchronization



Abnormal structure



Abnormal structure



Qualifiers

Symbol	Action
N	Non-stored
S	Set (Stored)
R	Reset
L	time Limited
D	time Delayed
P	Pulse
SD	Stored and time Delayed
DS	Delayed and Stored
SL	Stored and time Limited

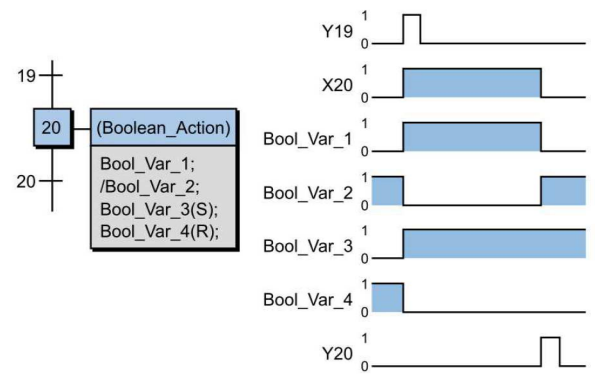


Figure 10-64. Example of a Boolean action.

Synchronization of operation

